

Ticketing & Train Journey Management

Case Study C1 · ArchiTrain · Focus Area 1

Abstract & Detailed ArchiMate Model

Group C1F1G_

Team: Member 1 · Member 2 · Member 3 · ...

TUM School CIT · 2026

Agenda

1

Use Case & Scope

ArchiTrain and our focus area F1

2

Abstract Model

The whole enterprise at a glance

3

Detailed Model — Focus Area F1

Business · Application · Technology

4

Reflection

Value · limitations · missing information · challenges

ArchiTrain: one architecture, three focus areas

A modern transit enterprise. Passengers **tap on / tap off** with a Go Card, mobile wallet or contactless bank card; the gate opens in under 400 ms — a single touch spanning hardware, networks, backend and payment.

F1

**Ticketing & Train
Journey Management**

Fare settlement & journey closure

◀ **OUR FOCUS**

F2

**Train Localization
& Scheduling**

GPS + odometry + track reference
points

F3

**Malfunction &
Disruption Management**

Threshold detection → human
handover

Key stakeholders & main services

Stakeholders

- Passenger (Business Actor)
- Customer Support Staff (Business Role)
- Operations Controller (Business Role)
- Train Operator (Business Role)
- Malfunction Management (Business Collaboration)

Main business services (→ serve Passenger)

- Ticketing & Fare Service [F1]
- Go Card Self-Service [F1]
- On-site Support Service [F1]
- Train Operation & Tracking Service [F2]
- Passenger Information Service [F2 / F3]

1 Abstract Model

Abstract model: three domains × three layers

Layer	F1 Ticketing & Journey	F2 Train Localization	F3 Disruption Mgmt
Business	Passenger · Customer Support Handle Journeys & Fares → Ticketing & Fare Service	Operations Controller · Train Operator Localize & Monitor Trains → Train Operation & Tracking	Malfunction Mgmt (collaboration) Handle Delays & Disruptions → Passenger Information Service
Application	Access Control Mgmt · JAMS Fare Management System · Website	LCCU Control Software Operational Control Center	Disruption Handling System (OCC + PIS + TMS)
Technology	Station Infrastructure · App Server Journey & Account Mgmt Backend	LCCU · Central Railway Backend Railway Radio	(shared backend) Private Network

Abstract level: one generalized process per domain · components without internal structure · devices summarized as nodes · communication as networks only.

2 Detailed Model — Focus Area F1

F1 spans business, application and technology

Business

Conduct Journey · Calculate Fare · Manage Go Card

Application

Access Control · JAMS · Fare Management · Website

Technology

NFC Reader · Gate · Backend · Private Network · Bank

Layers are linked by the two backbone relationships: **serving** and **realization**.

Business layer

Business layer — to be added

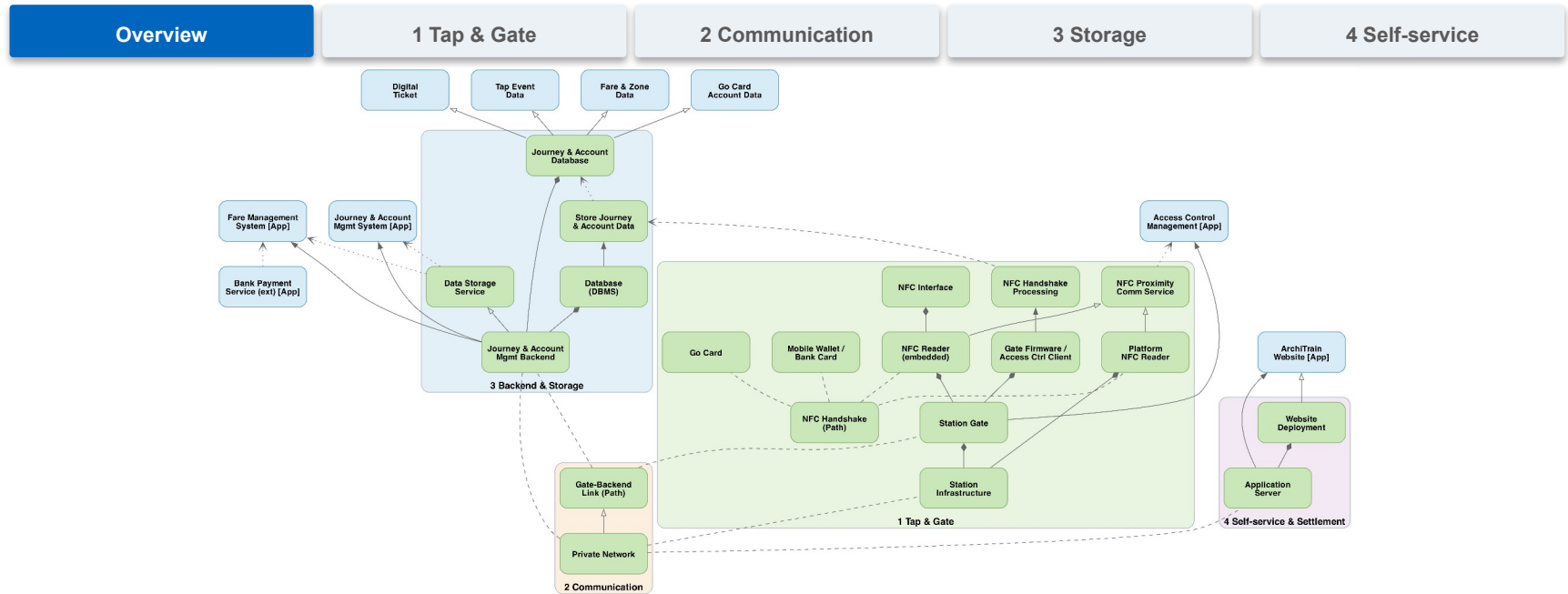
(pending teammate input: actors, processes, services, events)

Application layer

Application layer — to be added

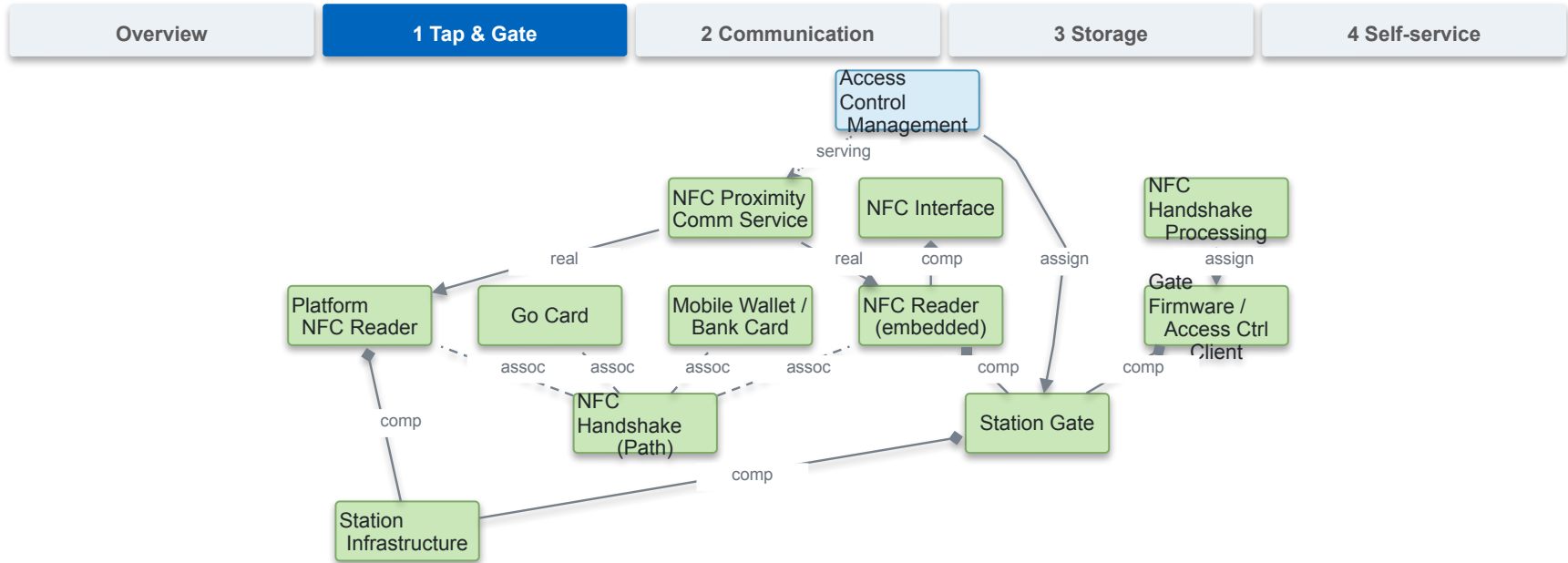
(components, application services, data objects)

Technology layer — overview & four perspectives



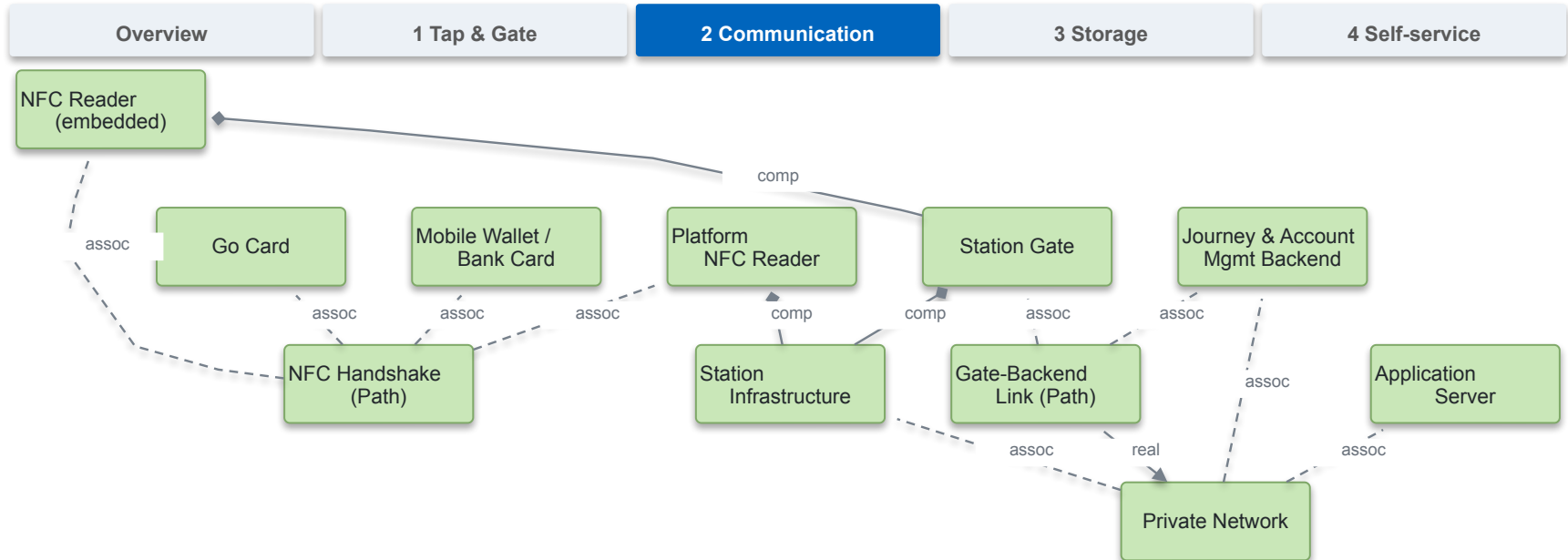
Full technology view grouped into four perspectives — examined one by one next.

Perspective 1 — Tap & gate access



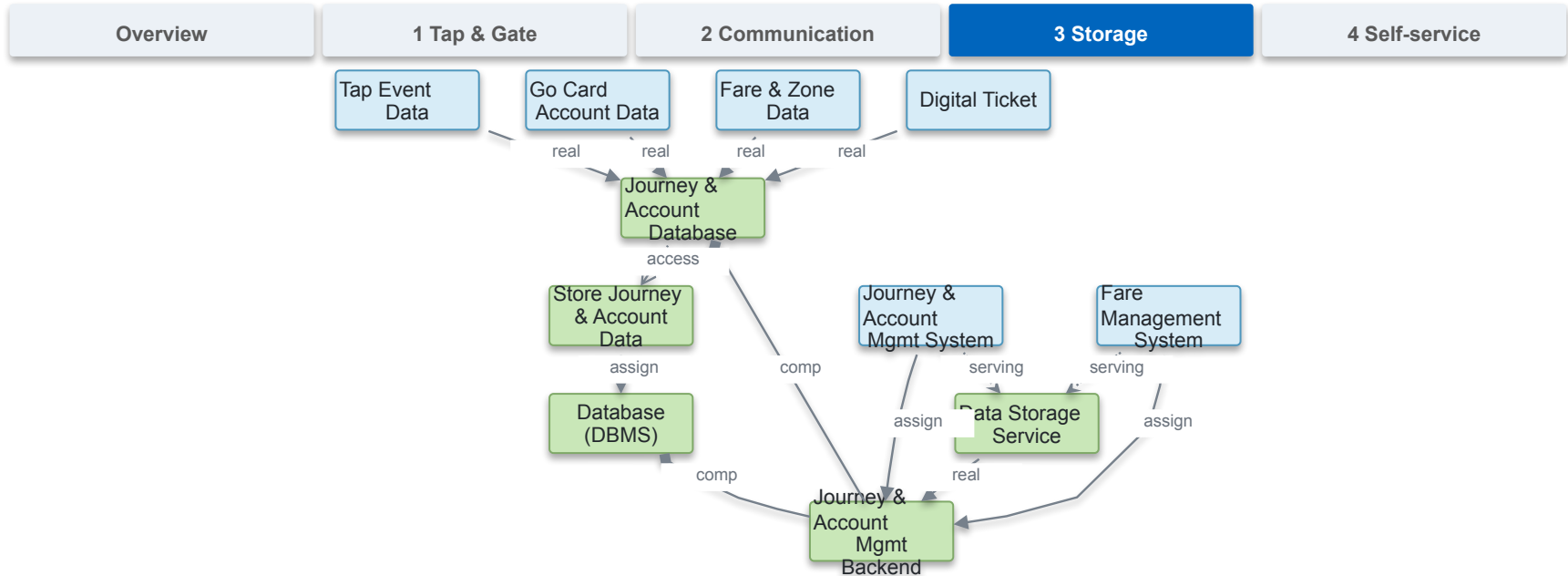
Passenger taps a Go Card / mobile wallet → NFC Handshake (Path) → reader; NFC Readers realize NFC Proximity Communication Service; Station Gate > Gate Firmware → NFC Handshake Processing; gate assigned to Access Control Management (< 400 ms).

Perspective 2 — Communication (pattern P01)



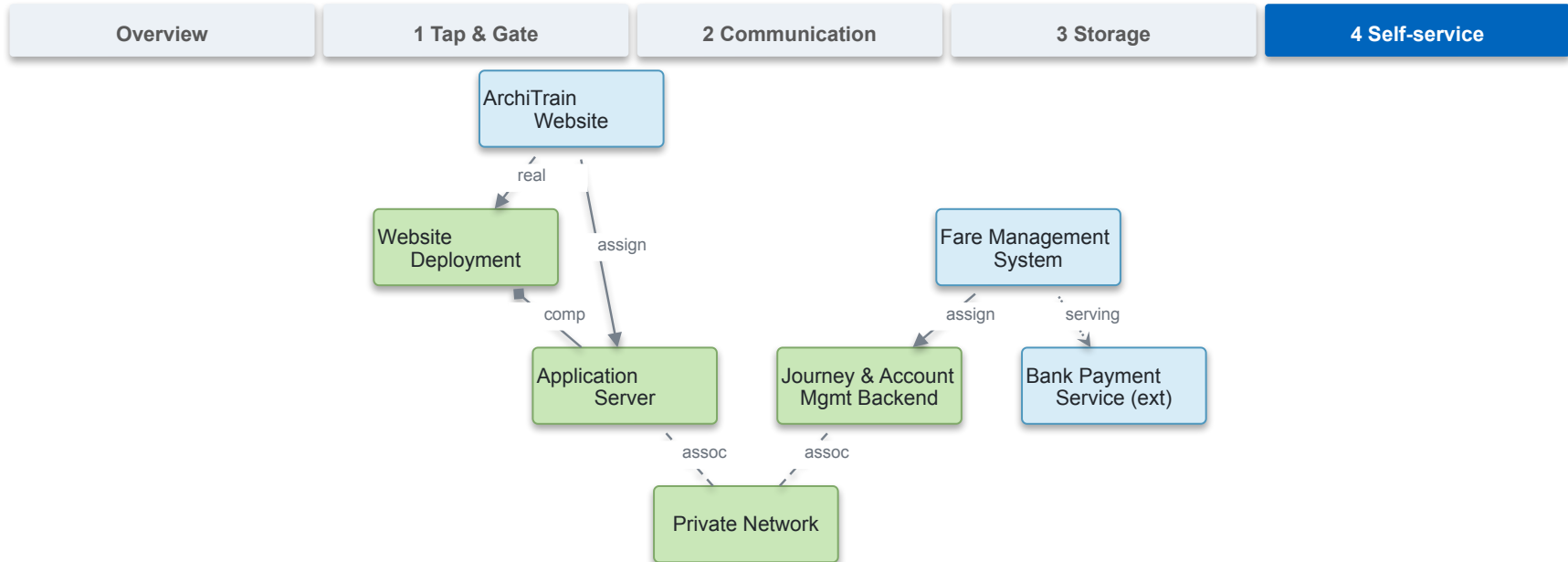
Card ↔ reader uses NFC Handshake (Path); Gate-Backend Link (Path) is realized by the Private Network (Communication Network) — course pattern P01; the network connects Station Infrastructure, the backend and the Application Server.

Perspective 3 — Backend storage & data



Journey & Account Mgmt Backend $\supset$ Database (DBMS) + Journey & Account Database (Artifact); the backend realizes the Data Storage Service serving JAMS & FMS; the artifact realizes Tap Event, Go Card Account, Fare & Zone and Digital Ticket data objects.

Perspective 4 — Self-service & bank settlement



Application Server > Website Deployment → realizes the ArchiTrain Website; the app server reads Go Card balances over the Private Network; Bank Payment Service (external) serves the Fare Management System for settlement.

3 Reflection

Reflection

Business value

[to be completed with the team]

Limitations

[to be completed with the team]

Missing information

[to be completed with the team]

Modeling challenges

[to be completed with the team]

Conclusion

- The abstract model covers the whole enterprise (F1 + F2 + F3) with structure and behaviour
- The F1 detailed model is developed across business, application and technology — communication explicitly modelled
- The vertical chain Node → Component → App Service → Business Process → Business Service → Passenger is derivable

Thank you — questions & discussion welcome.